

BAYESIAN SURVIVAL, LONGITUDINAL, AND JOINT MODELS WITH INLA - INTRODUCTION

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- 2017-2020: **Ph.D. in Public Health - Biostatistics, University of Bordeaux, France.** Development of joint models for oncology.
- 2021-2024: **Postdoctoral Fellow, KAUST University, Saudi Arabia.** Biostatistical models using the INLA methodology.
- Now: **Research Scientist at Inserm, University of Bordeaux, France.** Expanding and applying the framework to clinical & epidemiology research.

BEFORE WE START: WHAT TO EXPECT

- **This is a new course**, taught for the first time, built on recent research and materials published only months ago.
- New comes with trade-offs: you're getting state-of-the-art methods, but **not a course refined over decades of teaching**.
- **I'm an active researcher, not an experienced lecturer**, I develop these methods and I'll share them as clearly as I can.
- **Your feedback is welcome**. During the sessions, at the breaks, or anytime.
- One known bug in the instructor: **I speak (too) fast. Please tell me to slow down**. You'll be doing me and everyone else a favor.
- Questions are never interruptions. **If something isn't clear, it's probably my fault, so ask**.

- **Context:** Modern biomedical research where patients are followed over time.
- **Data:** Repeated biomarker measurements and clinically relevant events.
- **Objective:** We want to understand:
 - How does the disease evolve?
 - How do treatments alter this trajectory?
 - How do markers and clinical events inform each other?
- **Tools:** A unified framework to fit survival, longitudinal, and joint models efficiently using *INLA* and *INLAjoint*.

This course provides an overview of concepts and methodologies detailed in the new book:

“Bayesian survival, longitudinal and joint models with INLA”,
CRC Press (May 2026).

The book serves as a guide for researchers to tackle **complex biostatistical data** using a **flexible, computationally efficient**, and **unified Bayesian modeling framework**.

An online free version of the book is accessible at
https://rustand.fr/INLA_book.

1. **Event-time information** (death, progression, relapse, transplant or hospital readmission)
→ *terminal or recurrent, partially observed*
2. **Longitudinal information** (repeated biomarkers, imaging summaries, lab values or symptoms)
→ hierarchical structure, *different distributions (continuous, counts, binary, ordinal)*
3. **Joint modeling of 1. and 2.**
→ *challenging computations, various forms*
→ INLA

WHY NOT USE MCMC FOR EVERYTHING?

- Sampling-based Bayesian methods are general.
- The **computational burden** is one of the main reasons why rich joint models are still underused in practice.
- The book and this course exist partly because **recent INLA developments make these models much more accessible**.
- The models in this course quickly become **high-dimensional and hierarchical**. They involve **many random effects, several likelihoods** and **rich dependence structures**.
- In that setting, Bayesian inference gives a natural language for the uncertainty we actually care about. **INLA makes computations scalable** for large cohorts and very complex models.
- → We can go **beyond limitations of alternative methods**.

COMPUTATIONAL ADVANTAGE OF INLA

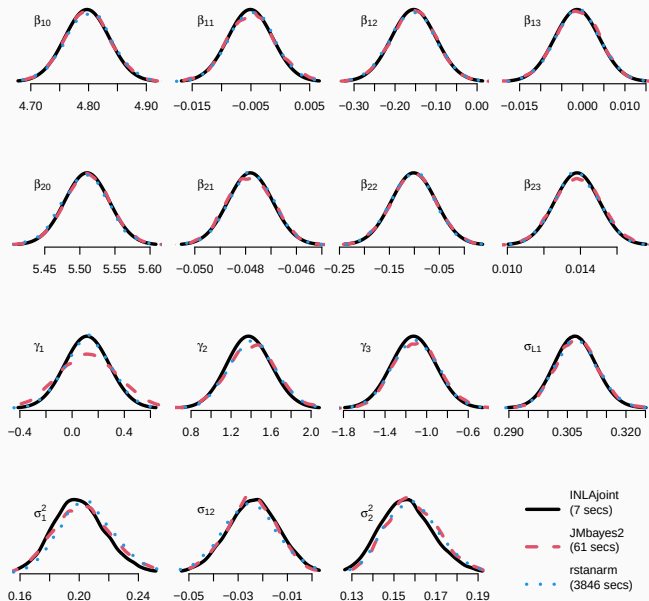
Computation Time Comparison (in seconds)

Model	INLA	MCMC (JAGS)	HMC (Stan)
Proportional Hazards	0.6	15	67
Multi-state	0.8	155	94
Joint Model	12	1754	949
Multivariate Joint Model	23	7105	9426

Danilo Alvares, Janet van Niekerk, Elias T. Krainski, Håvard Rue, Denis Rustand. *Bayesian survival analysis with INLA*. **Statistics in Medicine**. 2024; 43(20): 3975-4010. doi: 10.1002/sim.10160

Denis Rustand, Janet van Niekerk, Elias T. Krainski, Håvard Rue. *Joint modeling of multivariate longitudinal and survival outcomes with the R package INLAjoint*. **arXiv preprint** (2024) arXiv:2402.08335.

EXAMPLE: MULTIVARIATE JOINT MODEL



$$\pi(\boldsymbol{\theta} \mid \mathbf{D}) = \frac{\pi(\mathbf{D} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta})}{\pi(\mathbf{D})}$$

Component	Meaning in this course
$\pi(\boldsymbol{\theta} \mid \mathbf{D})$	Posterior distribution for fixed effects, random effects and hyperparameters
$\pi(\mathbf{D} \mid \boldsymbol{\theta})$	Likelihood from survival and longitudinal submodels
$\pi(\boldsymbol{\theta})$	Prior information and regularization
$\pi(\mathbf{D})$	Marginal likelihood, usually hard to compute exactly

INLA (Integrated Nested Laplace Approximations): A fast, accurate, and deterministic alternative to MCMC for a broad class of models known as Latent Gaussian Models (LGMs).

$$\pi(\boldsymbol{\chi}, \boldsymbol{\theta}) \propto \pi(\boldsymbol{D}|\boldsymbol{\chi}, \boldsymbol{\theta})\pi(\boldsymbol{\chi}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})$$

Instead of sampling, INLA directly calculates **highly accurate approximations** of the posterior marginals for each model parameter.

This approach is particularly efficient for the **hierarchical models** used in survival and longitudinal data analysis.

Rue, Håvard, Sara Martino, and Nicolas Chopin. *Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations*. Journal of the royal statistical society: Series b (statistical methodology) 71.2 (2009): 319-392.

The central idea from the book is that **a large family of useful models can be written in a common form**. For outcome k on subject i at time t ,

$$g\{E[D_{ik}(t)]\} = \eta_{ik}(t) = X_{ik}(t)^\top \beta_k + Z_{ik}(t)^\top b_{ik}.$$

This includes fixed effects, random effects, smooth functions and many familiar regression models.

INLA is designed for this latent Gaussian model family.

INLA stands for **Integrated Nested Laplace Approximations**.

1. **Approximates** the posterior distribution of the **hyperparameters**
2. **Approximates** the conditional posteriors of the **latent effects**
3. **Integrates** over the hyperparameters to recover **marginal posteriors**

INLA **avoids** long **sampling-based computation** by exploiting the structure of latent Gaussian models and **sparse precision matrices**.

INLAJOINT R PACKAGE: ACCESSIBLE IMPLEMENTATION

R-INLA is powerful but general. For the models in this course, writing everything directly in R-INLA would be possible but tedious and error-prone. **INLAjoint** provides a user interface specialized for **survival, longitudinal and joint modeling**.

It uses a “**building blocks**” syntax that is intuitive for users familiar with standard R modeling packages like **lme4** and **survival**.

```
joint(  
  formSurv = list(inla.surv(time, event) ~ trt,  
                  inla.surv(time2, event2) ~ trt,  
                  ...), # Survival submodels  
  formLong = list(Y ~ trt * time + (1 + time | id),  
                  Y2 ~ trt * time + (1 + time | id),  
                  ...), # Longitudinal submodels  
  assoc = list(c("CV", "CV"),  
               c("CV", "CV"),  
               ...), # Association structure  
  ...  
)
```

During this course, we are going to discuss **various modeling approaches** and we are going to show how all these models fit within the **same software framework**.

All the materials for the course will be uploaded on the following Github repository:

<https://github.com/DenisRustand/VIBASS9>